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DIRECTORATE OF TECHNICAL EDUCATION, GANDHINAGAR

L. D. COLLEGE OF ENGINEERING, AHMEDABAD

BACHELOR OF ENGINEERING (B.E.) — SEMESTER V

PROJECT MANAGEMENT MICRO-PROJECT REPORT

DESIGN AND IMPLEMENTATION OF A LOW-COST SUSTAINABLE SOLID WASTE SEGREGATION AND RECYCLING SYSTEM FOR CAMPUS WASTE MANAGEMENT

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CERTIFICATE OF APPROVAL

DIRECTORATE OF TECHNICAL EDUCATION, GUJARAT STATE

L. D. COLLEGE OF ENGINEERING, AHMEDABAD — 380015

This is to certify that the Project Management Micro-Project Report entitled:

"DESIGN AND IMPLEMENTATION OF A LOW-COST SUSTAINABLE SOLID WASTE SEGREGATION AND RECYCLING SYSTEM FOR CAMPUS WASTE MANAGEMENT"

is a bonafide record of work carried out by the following students of B.E. Semester V, Department of Information Technology, L. D. College of Engineering, Ahmedabad, during the academic session [2024–2025 / 2025–2026] in partial fulfillment of the requirements for the course of Project Management (Course Code: BE05000481).

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The report reflects original effort, rigorous project management methodologies (WBS, Gantt scheduling, RACI matrix, budget estimation, and risk analysis), and practical field-level investigation under faculty guidance. The technical prototype fabricated and pilot-tested is an original student model.

_____

Sign of Faculty Guide

Date: ___ / ___ / 202___

Place: Ahmedabad

ACKNOWLEDGEMENT

The successful conceptualization, fabrication, and pilot deployment of this Project Management Micro-Project would not have been possible without the invaluable guidance, constructive critique, and institutional support provided by esteemed faculty members, administrators, and campus staff.

We express our deepest sense of gratitude to Dr. [Principal's Name], Principal, L. D. College of Engineering, Ahmedabad, for fostering an academic environment that champions experiential learning, environmental sustainability, and interdisciplinary engineering projects.

We extend our heartfelt gratitude to Dr. / Prof. [HOD's Name], Head of the Information Technology Department, for his constant encouragement, administrative approvals, and for granting us permission to install the pilot segregation station within the department corridor.

We are profoundly indebted to our project guide and mentor, Prof. [Faculty Guide Name], Assistant Professor, Department of Information Technology, for his insightful guidance throughout the course of this micro-project. His mentorship in translating theoretical project management principles—such as Work Breakdown Structures, precedence diagramming, and risk mitigation matrices—into practical field execution was indispensable.

We also express our sincere appreciation to the Estate and Maintenance Department of LDCE for permitting the collection of discarded 20-litre paint drums and scrap timber from campus workshops, thereby embodying the spirit of circular waste upcycling. Our heartfelt thanks go to the campus housekeeping personnel and cafeteria staff whose daily collaboration made waste characterization audits, weighing, and trial segregation possible.

Finally, we commend each member of our student project team for their tireless dedication, mutual synergy, and unwavering commitment to engineering a cleaner, greener campus.

— B.E. Semester V (Information Technology) Project Team

L. D. College of Engineering, Ahmedabad

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Chapter 1
INTRODUCTION

Solid waste management represents one of the most critical civic, ecological, and institutional challenges confronting modern educational campuses across urban India. With rapid urban development, expanding academic infrastructure, and swelling student enrollments, educational institutions generate complex waste streams comprising paper, single-use food packaging, plastics, aluminium cans, botanical foliage, and cafeteria biodegradable scraps. In the absence of an organized source-segregation architecture, these diverse waste fractions become commingled at the point of disposal, transforming high-value recyclable commodities and nutrient-rich organic biomass into putrescible, contaminated refuse that ends up overburdening municipal dumpsites.

L. D. College of Engineering (LDCE), Ahmedabad, established in 1948 and affiliated with the Directorate of Technical Education, Gujarat State, spans a vibrant 59-acre campus hosting over 6,000 students, faculty, and administrative personnel across 14 engineering departments, laboratories, canteens, workshops, and student hostels. As a premier technological institution, LDCE has both the civic imperative and technical capability to pioneer sustainable, circular campus waste management models.

This micro-project, entitled "Design and Implementation of a Low-Cost Sustainable Solid Waste Segregation and Recycling System for Campus Waste Management", has been conceptualized, planned, and executed by B.E. Semester V Information Technology students as an experiential project management exercise. The overarching goal of the project is twofold:

- Technical Objective: To engineer and pilot-test a low-cost, decentralized solid waste segregation and recycling station utilizing repurposed materials, paired with an aerobic composting pilot and a digital waste audit log.
- Management Objective: To rigorously apply modern project management methodologies—including Project Charter formulation, Work Breakdown Structure (WBS), Gantt scheduling, RACI responsibility assignment, budget variance estimation, risk mitigation, and quality assurance—within an academic 8-week lifecycle and an austere budget ceiling of ₹1,700.

By bridging environmental engineering with information technology and formal project management, this initiative provides a scalable, empirical blueprint that transitions the college from passive, commingled waste disposal to an active, zero-waste circular campus model.

Chapter 2
BACKGROUND AND NEED FOR THE PROJECT

Rapid urbanization and institutional expansion in India have triggered an alarming escalation in the quantum and heterogeneity of municipal solid waste. Higher educational campuses, characterized by dense, transient student populations, extensive administrative operations, laboratory complexes, and food service centers, represent miniature municipalities. Nationally, the Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India, promulgated the landmark Solid Waste Management (SWM) Rules, 2016, which statutorily mandate all waste generators—including academic institutions, commercial establishments, and bulk waste generators—to segregate waste at source into three primary streams: biodegradable (wet), recyclable (dry), and domestic hazardous.

In Gujarat, the Gujarat Pollution Control Board (GPCB) and the Ahmedabad Municipal Corporation (AMC) have instituted stringent municipal bye-laws enforcing decentralized waste segregation under the national Swachh Bharat Mission (Urban) and Mission LiFE (Lifestyle for Environment). The city of Ahmedabad generates approximately 4,000 tonnes of municipal solid waste per day. Historically, a substantial portion of this unsegregated waste has accumulated at legacy dumpsites, notably the Pirana landfill, causing severe environmental degradation, subsurface groundwater contamination via toxic leachate, and uncontrolled methane emissions triggering spontaneous landfill fires.

A baseline reconnaissance survey conducted across L. D. College of Engineering revealed several compelling operational realities that establish the urgent need for this project:

- 1. Ubiquitous Commingled Disposal: Most departmental corridors, computer laboratories, and common areas are equipped with single, unsegregated open dustbins. Students and staff inadvertently dispose of half-eaten snacks, wet tea cups, paper assignments, plastic wraps, and discarded batteries into the same receptacle.
- 2. Value Destruction of Recyclables: When recyclable paper and clean cardboard come into contact with wet food scraps and sugary tea residues, the cellulose fibers degrade and become biologically contaminated, rendering hundreds of kilograms of high-grade paper unrecyclable and fit only for landfilling.
- 3. Sanitation and Public Health Risks: Open bins and unmanaged corner disposal points attract stray dogs, cats, rodents, and flies, creating persistent foul odors, visual blight, and potential hygienic hazards in high-footfall academic zones.
- 4. Behavioral Bottleneck: Despite high theoretical environmental awareness, the student body lacks clear visual cues, standardized color-coded receptacles, and hands-free hygienic disposal mechanisms at the immediate point of consumption.
- 5. High Capital Cost of Commercial Bins: Commercial segregation kiosks available from industrial vendors cost between ₹15,000 and ₹45,000 per unit, which is financially prohibitive for wide-scale collegiate adoption. Hence, there is a pressing need for a frugal, low-cost engineering solution (< ₹1,700) built from locally upcycled campus materials.

By tackling these five root causes, this micro-project demonstrates how disciplined project management, value-engineering, and student participation can convert an institutional waste liability into an ecological asset.

Chapter 3
PROBLEM STATEMENT

The formal problem statement addressed by this Project Management Micro-Project is formulated as follows:

KEY CONCEPT / FORMULA: FORMAL PROBLEM STATEMENT
"Educational facilities at L. D. College of Engineering generate substantial daily volumes of mixed solid waste that are currently collected in single, unsegregated bins. This lack of source segregation causes high rates of recyclable material contamination (>80%), eliminates on-site organic composting potential, produces corridor unhygienic conditions, and results in 100% of generated waste being routed to municipal landfills. Furthermore, the absence of quantitative waste tracking and high commercial bin costs prevent effective administrative intervention. There is an urgent need to engineer, pilot-test, and evaluate a low-cost (< ₹1,700), upcycled, multi-stream segregation station paired with a digital monitoring audit system to divert at least 80% of pilot waste from landfills."

To diagnose the systemic root causes leading to this problem, a Project Management Ishikawa (Fishbone) Cause-and-Effect Analysis was conducted across six dimensions:

- 1. Infrastructure / Equipment: Absence of multi-compartment color-coded receptacles; rusted single bins without foot pedals; lack of outdoor rain/sun protective canopies.
- 2. Process / Methods: Absence of a standard operating procedure (SOP) for campus housekeeping to maintain segregated collection; no scheduled tie-up with authorized AMC recyclers.
- 3. Materials / Waste Streams: Rapid discard of single-use food packaging, disposable cold drink PET bottles, and printed rough sheets during semester exams and laboratory submissions.
- 4. People / Behavioral: High general awareness among engineering students but poor disposal discipline; lack of immediate peer nudges; absence of Gujarati/Hindi visual infographics.
- 5. Measurement / Monitoring: Zero quantitative data on daily campus waste generation rates, composition percentages, or recyclable recovery value.
- 6. Environment / Climate: Ahmedabad's high ambient temperature (>40°C in summer) accelerating food waste fermentation and odor generation in unsegregated bins.

Chapter 4
OBJECTIVES OF THE PROJECT

The project objectives have been framed in strict accordance with the SMART (Specific, Measurable, Achievable, Relevant, and Time-bound) criteria foundational to Project Management Institute (PMI) standards.

SMART Criterion	Specific Project Target	Quantitative Metric / KPI
Specific	Design, fabricate, and deploy a low-cost, 3-stream segregation station using upcycled 20L commercial paint drums.	1 prototype unit installed in IT Dept corridor.
Measurable	Achieve a minimum waste diversion rate of 80% away from municipal landfill, and achieve >85% student segregation accuracy.	Diversion Rate ≥ 80% Segregation Accuracy ≥ 85%.
Achievable	Utilize locally available scrap materials from campus stores and execute entire fabrication at a cost under ₹1,700 within student capabilities.	Total Expenditure ≤ ₹1,700 Zero sophisticated workshop tools needed.
Relevant	Align campus waste handling with Solid Waste Management Rules 2016, CPCB norms, and Ahmedabad Municipal Corporation bylaws.	Full regulatory compliance with statutory 3-stream norms.
Time-bound	Execute all project phases—from charter initiation to pilot testing and final reporting—within an 8-week academic semester schedule.	Completion within exactly 8 academic weeks (56 days).

Chapter 5
SCOPE OF THE PROJECT

A fundamental principle of project management is precise scope definition to prevent 'scope creep'—the uncontrolled expansion of project activities without adjustments to time, cost, and human resources. Given that this micro-project is executed by a 5-member student team over an 8-week semester, a clear boundary has been established between in-scope deliverables and out-of-scope institutional elements.

IN-SCOPE DELIVERABLES (Committed)	OUT-OF-SCOPE BOUNDARIES (Excluded)
1. Fabrication of one (1) full-scale, 3-stream pilot waste segregation station using upcycled 20L paint drums.	1. Campus-wide civil construction or installation of hundreds of bins across all 14 engineering departments.
2. High-contrast trilingual (English, Gujarati, Hindi) visual infographics and universal recycling pictograms.	2. Procurement of expensive automated sensor-driven robotic sorting machinery or commercial balers.
3. Development of a lightweight digital waste audit log (Google Forms & Sheets) with daily weight tracking.	3. Complex enterprise ERP waste management software requiring paid cloud subscriptions.
4. Setup of a pilot aerobic composting demonstration bucket (5L) for canteen fruit and vegetable peels.	4. Commercial-scale biomethanation plants or mechanical mechanized composting drums.
5. Conduct of a 14-day waste audit in the IT Department corridor to measure daily weights and segregation accuracy.	5. Comprehensive chemical and microbiological laboratory leachate toxicity testing.
6. Formalized coordination with an authorized AMC scrap aggregator for dry recyclable handover.	6. Transportation or commercial processing of hazardous industrial chemicals or biomedical hospital waste.

Important Academic Disclaimer: This report documents the design, fabrication, deployment, and empirical evaluation of a single physical pilot prototype station installed in the IT Department corridor of L. D. College of Engineering. References to campus-wide waste collection or long-term institutional expansion represent strategic recommendations, not claims of completed campus-wide civil installation.

Chapter 6
TYPES AND SOURCES OF SOLID WASTE GENERATED ON CAMPUS

To formulate an effective waste management system, a comprehensive baseline waste audit was conducted to identify the sources, functional areas, and specific typologies of solid waste generated across L. D. College of Engineering.

Campus Zone / Source	Primary Waste Types	Composition & Characteristics	Stream Allocation
IT & Engineering Labs, Classrooms, Admin Offices	Printouts, assignment pages, old notebooks, exam answer sheets, pen refills, corrugated cardboard packaging.	Dry, clean cellulose fibers, highly recyclable if kept free of moisture and grease.	Blue Stream (Dry Recyclable)
College Canteen, Food Kiosks, Mess	Leftover rice, roti, vegetable curries, fruit peels, tea leaves, used tea paper cups, soiled paper napkins.	High moisture (>65%), rapidly biodegradable, putrescible, high potential for composting.	Green Stream (Wet Organic)
Student Hostels & Residential Quarters	Plastic shampoo bottles, biscuit/chip multi-layer wrappers, cardboard soap boxes, worn textiles, dust sweepings.	Heterogeneous mix; requires segregation into recyclables vs. non-recyclable wrappers.	Blue & Black Streams
Campus Grounds, Lawns & Botanical Zones	Fallen leaves (neem, gulmohar), dry twigs, trimmed lawn grass, garden soil cuttings.	Carbon-rich organic biomass (brown waste); excellent structural bulking agent for composting.	Green Stream (Brown Carbon Fraction)
Hardware Labs & Campus Health Center	Dry cell batteries (AA/AAA), broken CFL tubes, toner cartridges, sanitary napkins, adhesive bandages.	Domestic hazardous and sanitary waste; requires safe, non-contaminating containment.	Red / Yellow Stream

Based on our 3-day baseline audit conducted prior to system deployment, the campus waste generated across academic corridors exhibits the following average gravimetric distribution:

- Wet Organic / Biodegradable Waste: 52.6% (dominated by canteen food scraps, fruit rinds, and tea grounds).
- Dry Recyclable Waste: 35.9% (clean paper 18.2%, PET plastic bottles 11.4%, cardboard 4.1%, metal cans 2.2%).
- Non-Recyclable Rejects / Inerts: 11.5% (metallized plastic chip pouches, soiled wrappers, sweepings dust).

Chapter 7
PROPOSED SUSTAINABLE WASTE MANAGEMENT SYSTEM

The proposed system establishes a low-cost, decentralized, circular solid waste segregation and recycling architecture engineered specifically for the operational dynamics of an engineering college campus. The core philosophy of the system is source segregation—preventing waste fractions from ever mingling, thereby preserving the economic value of recyclables and the biological purity of organic compostable matter.

Figure 2.1: Integrated Campus Solid Waste Management Lifecycle Flowchart

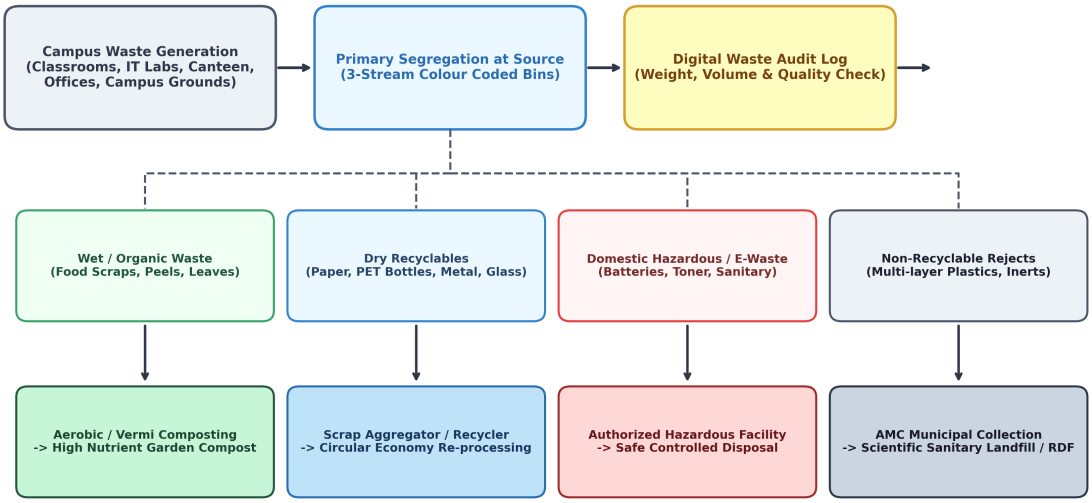


Figure 2.1: Integrated Campus Solid Waste Management Lifecycle Flowchart

7.1 Four-Stream Segregation Protocol

In strict compliance with the Solid Waste Management Rules, 2016, and CPCB institutional guidelines, waste on campus is classified into four distinct streams, each assigned a standardized color code, aperture geometry, and visual infographic:

Figure 3.1: Campus Waste Segregation Protocol & Classification Guide

(In Accordance with Solid Waste Management Rules, 2016 & CPCB Guidelines)



Figure 3.1: Campus Waste Segregation Protocol & Classification Guide

- 1. Green Stream (Wet / Biodegradable Waste): Dedicated exclusively to organic, putrescible matter generated primarily from cafeteria leftovers, tiffin boxes, fruit peels, tea bags, and fallen campus garden foliage. The collected organic waste is transferred daily to the campus aerobic composting unit. Key Rule: No plastic bags, foils, or non-biodegradable wrappers.
- 2. Blue Stream (Clean Dry Recyclables): Dedicated to high-grade recyclable commodities including paper printouts, notebooks, exam sheets, flattened corrugated cardboard cartons, rinsed PET beverage bottles, and aluminium soft-drink cans. Key Rule: All beverage containers must be emptied and crushed before deposition to optimize volumetric capacity.
- 3. Red / Yellow Stream (Domestic Hazardous & E-Waste): A specialized secure sub-stream positioned primarily in hardware laboratories and departmental staff rooms to capture dry cell batteries, printer toners, spent fluorescent/LED lamps, broken lab glassware, and sanitary waste wrapped in paper. Key Rule: Prevents heavy metals and bio-contaminants from polluting common recycling streams.
- 4. Black Stream (Non-Recyclable Rejects / Inerts): Reserved strictly for contaminated, non-recyclable residual inerts such as multi-layered metallized snack packets (chips wrappers), wax-coated papers, soiled sweeping dust, and broken ceramics. This fraction represents the sole stream designated for Ahmedabad Municipal Corporation (AMC) landfill collection.

7.2 Waste Processing & Routing Decision Logic

To ensure that every item deposited reaches its highest ecological value, a deterministic decision workflow governs campus waste routing:

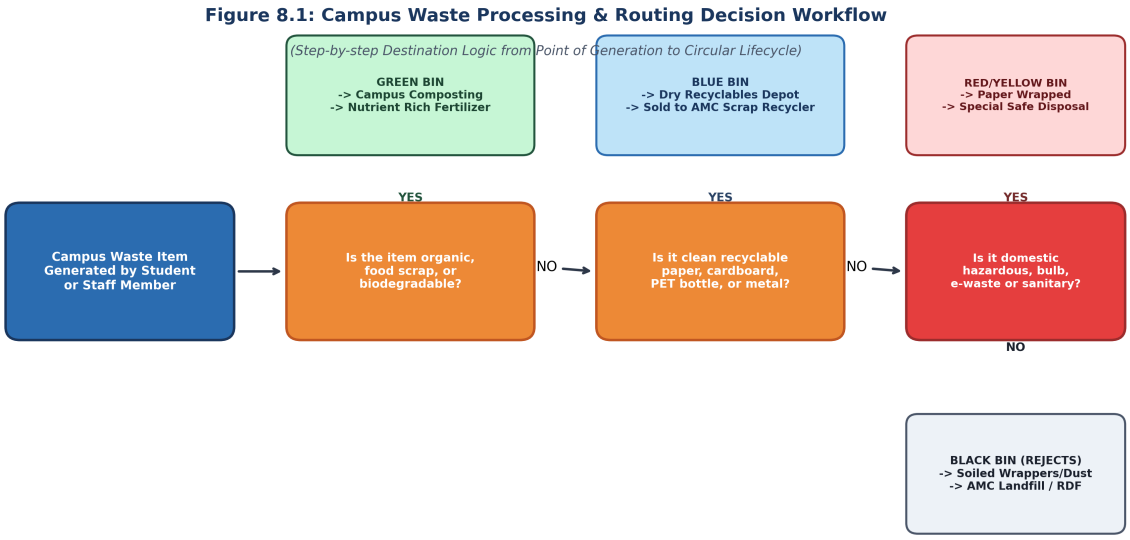


Figure 8.1: Campus Waste Processing & Routing Decision Workflow

7.3 In-Situ Aerobic Composting Mechanism

For the biodegradable organic fraction, the system employs decentralized aerobic composting. A pilot 20-litre perforated composting unit was prepared with aeration holes (8 mm diameter drilled circumferentially) and an internal mesh drainage base to collect excess liquid leachate. Composting operates on the principle of microbial decomposition maintained at an optimal Carbon-to-Nitrogen (C:N) ratio of approximately 25:1 to 30:1:

- Nitrogen-rich 'Green' inputs: Cafeteria vegetable peels, fruit rinds, leftover food (moisture: ~65%, high N).
- Carbon-rich 'Brown' bulking agents: Crushed dry neem/gulmohar leaves collected from campus lawns and shredded unbleached cardboard packaging (provides structural porosity and absorbs excess moisture).
- Microbial Inoculum: A starter spray of diluted sour buttermilk (chaas) and local fertile soil rich in indigenous composting bacteria and actinomycetes was added to accelerate mesophilic decomposition without foul odor.

Chapter 8
MATERIALS, EQUIPMENT, AND RESOURCES REQUIRED

A pivotal principle of this micro-project is extreme cost-efficiency and circularity. Wherever feasible, materials were salvaged, repurposed, or donated from campus maintenance stores, strictly limiting purchases to essential paints, fasteners, and safety PPE.

Resource / Material Description	Source / Nature	Quantity / Specifications	Functional Application
Discarded 20L Plastic Buckets	Upcycled (Campus Stores)	3 Units (Heavy-Duty HDPE)	Core receptacles for 3-stream segregation station
Scrap Timber Pallet / Angle Iron	Upcycled (Workshop Scrap)	1 Assembly (4 ft x 1.5 ft)	Elevated base skid to prevent water contact
Discarded 5L Plastic Bucket	Upcycled (Cafeteria Scrap)	1 Unit (Perforated)	Pilot aerobic composting demonstration unit
Scrap GI Rods & Cycle Brake Wire	Upcycled (Campus Workshop)	2 Rods (8 mm), 1 Cable	Mechanical foot-pedal lid lifting linkage
Synthetic Enamel Paint (Green, Blue, Black)	Purchased (Local Hardware)	500 ml Green, 500 ml Blue, 200 ml Black	Weatherproof, color-coded protective bin coating
Anti-Rust Primer & Turpentine Thinner	Purchased (Local Hardware)	250 ml Primer, 500 ml Thinner	Surface preparation and paint dilution
Computer-Cut Vinyl Lettering & Stencils	Purchased (Signage Shop)	3 Sets (Trilingual Text & Icons)	Bilingual labels (English & Gujarati) + recycling logos
PP Corrugated Fluted Sheet (3x2 ft)	Purchased (Stationery)	1 Sheet (Sunpack Yellow)	Overhead instructional canopy & splash guard
Foot-Pedal Hardware & Fasteners	Purchased (Hardware Store)	2 Hinges, Return Spring, Nuts/Bolts	Hands-free foot lever fabrication
Digital Hanging Luggage Scale	Purchased (Electronics)	1 Unit (50 kg / 10g precision)	Daily gravimetric waste audit and KPI logging
Safety Gloves & N95 Dust Masks	Purchased (Medical Store)	4 Pairs Nitrile Gloves, 10 Masks	Occupational PPE for waste handling and sorting
A4 Awareness Posters & QR Laminates	Purchased (Print Shop)	10 Colour Prints (Laminated)	Educational signage and digital audit scan placards
Microbial Culture & Coco-Peat	Purchased (Agri-Nursery)	2 kg Coco-Peat Block + Culture	Composting starter and carbon bulking medium

Chapter 9
PROJECT DESIGN AND WORKING METHODOLOGY

The project design embodies human-centered design (ergonomics), behavioral psychology (nudging theory), and hygienic engineering. To ensure wide adoption among engineering students and faculty, the station layout addresses common friction points identified during our initial campus survey.

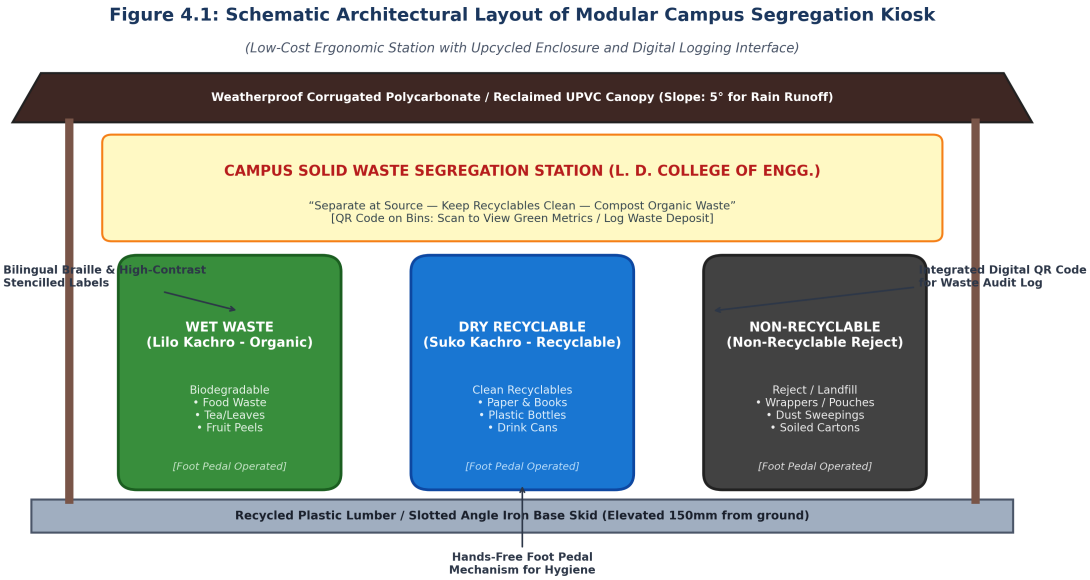


Figure 4.1: Schematic Architectural Layout of Modular Campus Segregation Kiosk

9.1 Ergonomic and Physical Engineering Specifications

- 1. Optimal Aperture Height: Bin rim height is positioned at exactly 850 mm from ground level, allowing effortless waste deposition for individuals ranging from 150 cm to 185 cm in height without bending.
- 2. Hands-Free Foot-Pedal Mechanism: To overcome student hesitation regarding touching potentially dirty bin lids, each bin is actuated by a mechanical foot pedal. Depressing the foot pedal raises the lid through a lever linkage; releasing it automatically drops the lid via gravity and return spring.
- 3. Visual Nudge Signage: Mounted at an eye-level elevation of 1.5 meters, a bright yellow canopy display features high-contrast text and realistic item illustrations in both English and Gujarati, removing any cognitive ambiguity at the moment of discard.
- 4. Modular Base Elevation: The station is elevated 150 mm on a timber/slotted angle iron skid with anti-slip rubber pads, protecting the bins from waterlogging during monsoon rains and simplifying corridor floor sweeping.

9.2 Working Methodology: The PDCA Quality Cycle

The technical execution follows the classical Plan-Do-Check-Act (PDCA) quality framework advocated by Edwards Deming, ensuring structured continuous improvement throughout the project lifecycle:

- Phase 1: PLAN (Weeks 1–3): Conducted stakeholder interviews with campus administration, housekeeping supervisors, and students. Carried out a 3-day baseline waste characterization audit to estimate daily volumes and contamination rates. Formulated project charter and technical specifications.
- Phase 2: DO (Weeks 4–6): Salvaged discarded 20L paint drums from campus stores. Fabricated the physical 3-stream pilot station, applied multi-coat enamel finishes, attached foot pedals, and installed bilingual signage. Deployed the station in the high-footfall ground floor corridor of the IT Department.
- Phase 3: CHECK (Week 7): Conducted a rigorous 14-day daily waste audit. Weighed each stream gravimetrically every morning at 10:00 AM using the calibrated digital scale. Logged data into the cloud-based Google Sheets audit tracker. Monitored contamination instances and composter temperature/moisture.
- Phase 4: ACT (Week 8): Analyzed audit logs, calculated key performance indicators (Landfill Diversion Rate, Segregation Accuracy %), adjusted compost bulking ratios to eliminate moisture, and established a formalized recyclable collection channel with an authorized AMC scrap partner.

9.3 Digital Waste Monitoring & Analytics Dashboard

Capitalizing on the team's Information Technology engineering background, a lightweight digital waste monitoring system was developed. Each bin features a weather-resistant QR code placard. Scanning the QR code directs students and housekeeping staff to a Google Form audit portal.

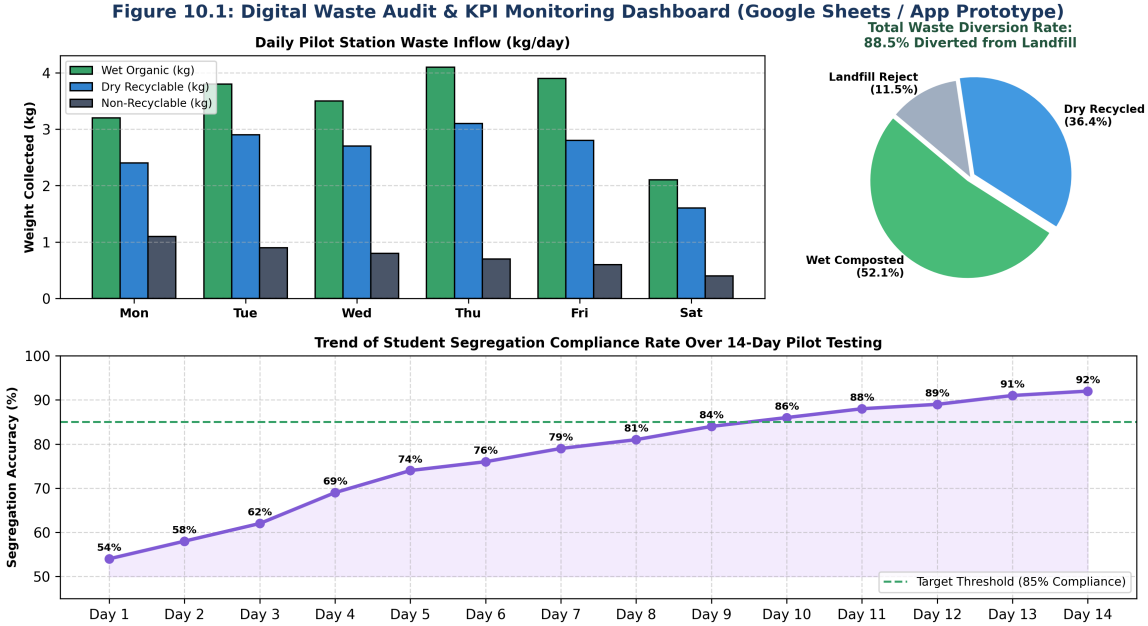


Figure 10.1: Digital Waste Audit & KPI Monitoring Dashboard

KEY CONCEPT / FORMULA: MATHEMATICAL FORMULAS FOR PERFORMANCE MONITORING

To evaluate system efficiency rigorously without unsubstantiated claims, the following empirical formulas are programmed into the analytics dashboard:

1. Landfill Waste Diversion Rate (%):

Diversion Rate = [(Mass of Composted Wet Waste + Mass of Recycled Dry Waste) / Total Mass of Waste Collected] x 100

2. Stream Segregation Accuracy (%):

Accuracy = [1 - (Mass of Contaminants Detected in Stream / Total Mass of Stream)] x 100

3. Recyclable Recovery Rate (%):

Recovery Rate = [Mass of Clean Recyclables Handed to Recycler / Estimated Total Recyclables Generated] x 100

Chapter 10
STEP-BY-STEP IMPLEMENTATION / CONSTRUCTION PROCESS

The physical fabrication of the pilot waste segregation station was executed systematically by the student project team in the college workshop over an intensive 10-day period. The entire construction process was documented to ensure seamless reproducibility by other departments across LDCE.

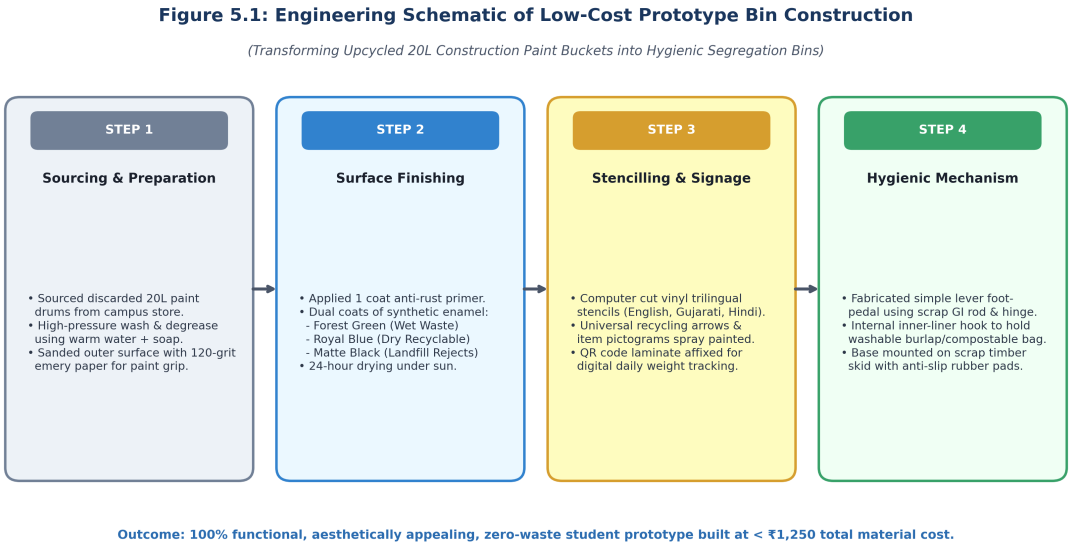


Figure 5.1: Engineering Schematic of Low-Cost Prototype Bin Construction

Phase 1: Salvage, Degreasing, and Mechanical Preparation

Three discarded 20-litre commercial plastic paint drums were retrieved from the campus maintenance storage yard. The containers were subjected to a rigorous multi-stage cleaning process: scraped free of residual dried paint using putty knives, degreased with warm detergent solution, and high-pressure water washed. After sun drying, eight 6-mm ventilation holes were drilled around the top rim of the wet waste bin to prevent anaerobic gas buildup.

Phase 2: Surface Abrasion and Dual-Coat Enamel Application

Because untreated high-density polyethylene (HDPE) plastic has low surface energy, the outer surfaces were abraded uniformly using 120-grit emery sandpaper to create mechanical anchoring profiles. A single base coat of synthetic primer was applied and cured for 12 hours. Next, two heavy coats of quick-drying synthetic enamel were applied:

- Forest Green Enamel: Wet / Biodegradable Waste Receptacle (Coded according to SWM Rules 2016).
- Royal Blue Enamel: Clean Dry Recyclable Waste Receptacle.
- Matte Jet Black Enamel: Non-Recyclable Rejects / Landfill Receptacle.

Phase 3: Stencilling, Bilingual Signage, and QR Placement

Computer-cut vinyl stencils were utilized to spray-paint high-contrast white lettering on each bin. Each receptacle was labeled in both English and Gujarati (e.g., 'WET WASTE / LILO KACHRO'), accompanied by universal recycling arrows and iconography depicting accepted items. A laminated QR code tag was affixed to the upper right rim with waterproof epoxy resin.

Phase 4: Mechanical Foot-Pedal Linkage & Skid Assembly

A simple, resilient foot-pedal lever mechanism was fabricated using salvaged 8-mm galvanized iron rods and heavy-duty steel butt hinges mounted to the timber skid. Depressing the foot pedal tensions a steel pull cable routed through guide eyes, cleanly tilting open the hinged bucket lid without manual contact.

Phase 5: Pilot Aerobic Composter Preparation

A 5-litre plastic container was perforated with aeration holes, fitted with a coarse mesh false bottom, and primed with a 30 mm base layer of dried leaves and coco-peat. Vegetable cuttings and tea waste collected from the canteen were deposited in stratified layers and inoculated with diluted fermented buttermilk.

Phase 6: Corridor Deployment and Housekeeping Orientation

The assembled pilot station was transported to the ground floor foyer of the IT Department building. An orientation session was conducted with the departmental housekeeping staff to explain daily collection protocols, separate bag handling, and digital weighing procedures.

Chapter 11
PROJECT MANAGEMENT PLAN

To transition this initiative from an informal student activity to a disciplined engineering deliverable, a comprehensive Project Management Plan was developed aligning with the PMBOK (Project Management Body of Knowledge, 7th Edition) framework. The plan encompasses six integrated subsidiary management plans:

- 1. Scope Management Plan: Defines all deliverables required for 1 functional pilot station, digital dashboard, and 14-day audit. Changes to scope require formal review by the Project Lead and Faculty Guide.
- 2. Schedule Management Plan: Establishes an 8-week semester schedule using critical path method (CPM) logic. Progress tracked via weekly milestone reviews and Gantt chart variance tracking.
- 3. Cost Management Plan: Enforces strict cost control under a total budget ceiling of ₹1,700 INR. All expenses documented with GST bills; 10% allocated as contingency reserve.
- 4. Quality Management Plan: Sets measurable quality acceptance criteria: minimum 80% waste diversion, >85% segregation accuracy, zero odor within 2 meters of composter, and durable paint adhesion.
- 5. Risk Management Plan: Maintains an active Risk Register tracking 10 prioritized risks with assigned mitigation strategies and contingency triggers.
- 6. Stakeholder Engagement Plan: Categorizes stakeholders (Students, Faculty, Housekeeping, HOD, AMC Recyclers) into a Power-Interest grid, with tailored communication channels for each group.

Chapter 12
WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) decomposes the entire project into manageable, hierarchical work packages, ensuring that 100% of the project scope is captured without unnecessary overlap (the 100% Rule).

Figure 6.1: Work Breakdown Structure (WBS) for Campus Solid Waste Project

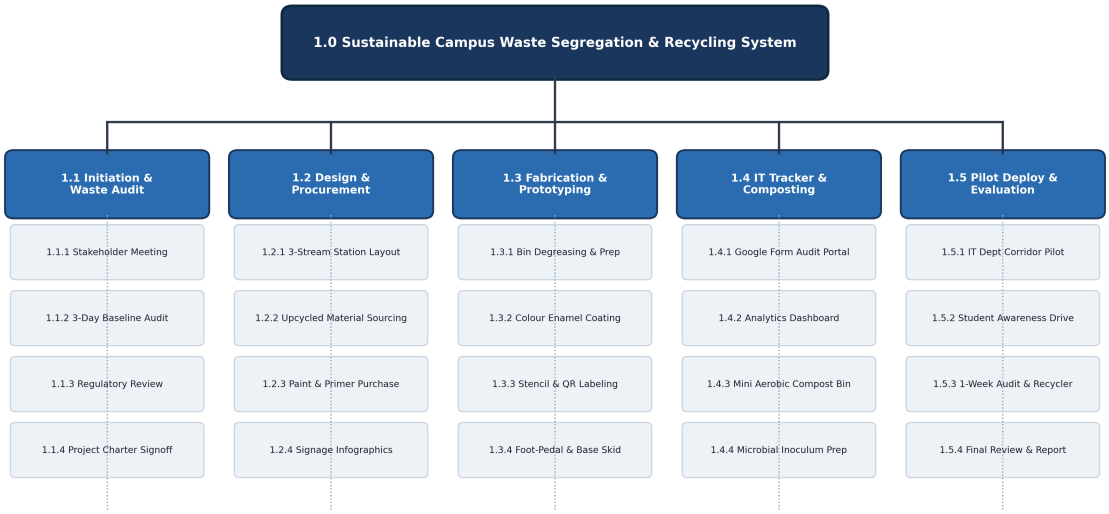


Figure 6.1: Work Breakdown Structure (WBS) Hierarchical Chart

12.1 Detailed WBS Dictionary

The WBS Dictionary defines each terminal work package, specifying ownership, scope details, deliverables, and formal acceptance criteria:

WBS Code	Work Package Name	Assigned Lead & Key Activities	Acceptance Criteria / Deliverable
1.1.1	Stakeholder Consultation	Team Lead: Consult HOD, Estate Officer & Housekeeping Supervisor.	Written corridor deployment permission signed.
1.1.2	Baseline Waste Audit	Waste Audit Lead: 3-day gravimetric measurement of unsegregated bins.	Baseline dataset (kg/day & % composition) completed.
1.1.3	Regulatory Compliance Review	QA Lead: Review SWM Rules 2016, CPCB and AMC municipal guidelines.	Regulatory compliance checklist approved by guide.
1.2.1	Station Layout & Sizing	Fabrication Lead: Ergonomic design, CAD dimensions, aperture spacing.	Dimensioned mechanical layout drawings finalized.
1.2.2	Material Sourcing & Scrap	Procurement Lead: Retrieve 3 paint	All raw scrap materials inspected

	Salvage	buckets and timber scrap from estate store.	and stored in workshop.
1.3.1	Bin Surface Prep & Painting	Fabrication Lead: Degreasing, sanding, primer and dual enamel coating.	Scratch-resistant, uniform color-coated buckets ready.
1.3.2	Foot-Pedal & Base Fabrication	Fabrication Lead: Build lever pedal, hinge assembly, and timber base.	Hands-free lid opening tested >50 cycles successfully.
1.4.1	Digital Audit Tool Setup	IT Lead: Develop Google Form, QR codes, and Sheets analytics formulas.	Live QR code scanning functional on mobile devices.
1.4.2	Pilot Composter Setup	Compost Lead: Perforate 5L container, prepare bed, mix canteen peels.	Active aerobic composting initiated with zero foul odor.
1.5.1	Corridor Deployment & Pilot Audit	Full Team: 14-day daily weighing, student awareness nudges, recycler pickup.	Completed 14-day audit dataset and formal final report.

Chapter 13
PROJECT SCHEDULE AND GANTT CHART

Project activities were sequenced and scheduled over an 8-week semester timeframe (56 calendar days). Dependencies between tasks were established using Finish-to-Start (FS) and Start-to-Start (SS) logic to identify the Critical Path.

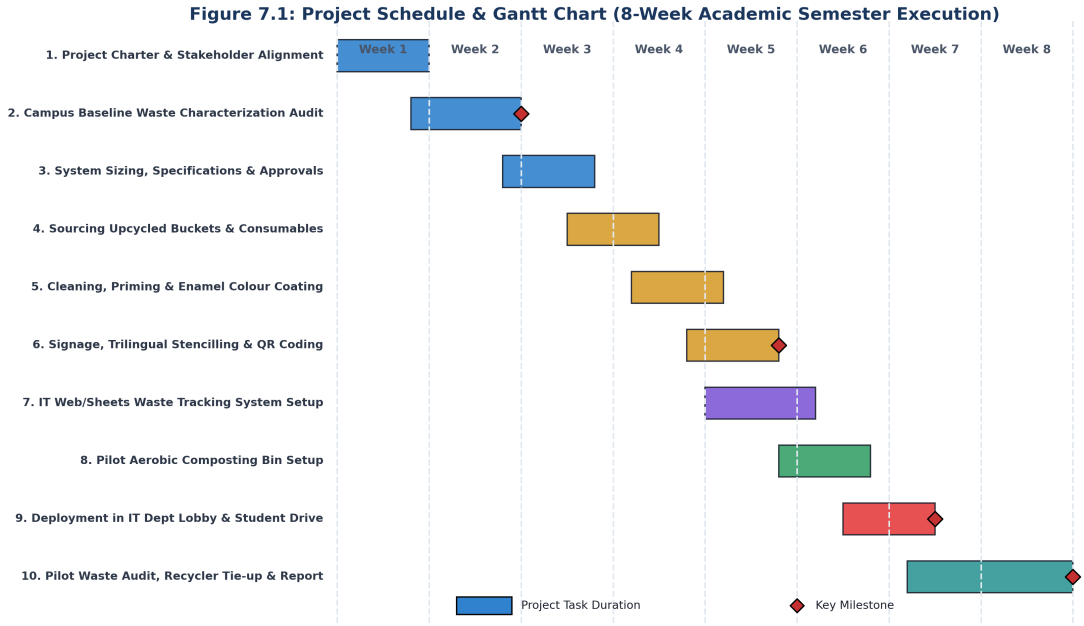


Figure 7.1: Project Schedule & Gantt Chart (8-Week Academic Semester)

Task ID	Activity / Work Package Description	Predecessors	Duration	Start	Finish
T-01	Project Charter Formulation & Guide Signoff	None	5 Days	Day 1	Day 5*
T-02	Campus Baseline Waste Characterization Audit	T-01	7 Days	Day 6	Day 12*
T-03	System Specifications & Department Approvals	T-02	5 Days	Day 13	Day 17
T-04	Scrap Material Sourcing & Hardware Purchases	T-03	5 Days	Day 18	Day 22*
T-05	Bin Degreasing, Priming & Enamel Painting	T-04	6 Days	Day 23	Day 28*
T-06	Stencilling, Vinyl Labelling & Foot-Pedal Assembly	T-05	5 Days	Day 29	Day 33*
T-07	IT Digital Tracking System Setup (Google Forms/Sheets)	T-03	6 Days	Day 25	Day 30
T-08	Pilot Aerobic Composter Bin Setup & Inoculum	T-05	4 Days	Day 30	Day 33
T-09	Deployment in IT Dept Lobby & Student	T-06, T-07	4 Days	Day 34	Day 37*

	Orientation				
T-10	14-Day Pilot Audit, Data Analytics & Recycler Tie-Up	T-09	14 Days	Day 38	Day 51*

**Indicates Critical Path activities (Total Duration = 51 Days + 5 Days buffer for final documentation review = 56 Days / 8 Academic Weeks). Any delay in critical path activities directly postpones the final project completion date.*

Chapter 14
TEAM ROLES AND RESPONSIBILITIES

A successful project management execution relies on unambiguous role allocation and accountability. A 5-member organizational structure was instituted, with each student assigned clear primary responsibilities and backup support roles.

14.1 RACI Responsibility Assignment Matrix

The RACI Matrix defines the relationship between team members and key project deliverables across four levels of participation: Responsible (R), Accountable (A), Consulted (C), and Informed (I):

Major Project Deliverable	Lead 1 (PM)	Lead 2 (Fab)	Lead 3 (Waste)	Lead 4 (IT)	Lead 5 (QA)
1. Project Charter & Scope Baseline	A / R	C	C	C	C
2. Baseline Campus Waste Audit	I	C	A / R	R	C
3. Scrap Sourcing & Procurement	C	A / R	C	I	R
4. Station Fabrication & Painting	I	A / R	C	I	R
5. Foot-Pedal Linkage Assembly	I	A / R	I	I	C
6. Bilingual Signage & QR Placards	C	R	C	A / R	C
7. IT Google Forms & Analytics Tool	I	I	C	A / R	C
8. Aerobic Composter Bin Setup	I	C	A / R	I	C
9. 14-Day Pilot Data Collection	A	C	R	R	R
10. Recycler Handover & Final Report	A / R	C	R	R	A / R

Chapter 15
RESOURCE PLANNING AND COST ESTIMATION

Cost management is paramount in value-engineering. The budget was structured into two clear accounts: Salvaged/Repurposed Assets (zero acquisition cost) and Direct Purchased Consumables. The financial objective was to deliver a complete, durable station under ₹1,700 INR.

Item	Item Description & Specifications	Category	Rate (₹)	Total (₹)
1	3x 20L Plastic Commercial Paint Buckets	Upcycled (Estate)	₹0 (Free)	₹0.00
2	Timber Pallet Scrap / Angle Iron Base Skid	Upcycled (Workshop)	₹0 (Free)	₹0.00
3	1x 5L Plastic Bucket for Pilot Composter	Upcycled (Canteen)	₹0 (Free)	₹0.00
4	Scrap GI Rods & Bicycle Cable for Foot Lever	Upcycled (Workshop)	₹0 (Free)	₹0.00
5	Synthetic Enamel Paint (Green, Blue, Black)	Purchased	Lump sum	₹360.00
6	Painting Brushes (2-in & 1-in) + Emery Paper	Purchased	Set of 3	₹110.00
7	Computer-Cut Vinyl Lettering & Stencils	Purchased	3 Sets	₹180.00
8	PP Fluted Canopy Sheet (3x2 ft) + Fasteners	Purchased	1 Assembly	₹140.00
9	Foot-Pedal Hardware (Hinges, Springs, Screws)	Purchased	Hardware Kit	₹120.00
10	Digital Hanging Luggage Scale (50 kg / 10g)	Purchased	1 Unit	₹240.00
11	Safety Nitrile Gloves & Protective Masks	Purchased	Medical Pack	₹160.00
12	Laminated A4 Informational Posters & QR Placards	Purchased	10 Sheets	₹120.00
13	Coco-Peat Brick (2kg) & Compost Microbial Starter	Purchased	Pack	₹90.00
--	SUBTOTAL DIRECT PURCHASES	--	--	₹1,520.00

Project Cost Baseline Summary:

- Subtotal Purchased Materials: ₹1,520.00
- Project Management Contingency Reserve (10% of purchases): ₹152.00
- Total Authorized Project Budget: ₹1,672.00 (Approximately ₹1,670 INR)
- Commercial Equivalent Cost: ₹18,500.00 (Resulting in a 91% cost savings via circular upcycling!)

Chapter 16
RISK IDENTIFICATION AND MITIGATION PLAN

Proactive risk management is indispensable for safeguarding project delivery, operational safety, and stakeholder adoption. A quantitative Risk Register was compiled during the planning phase, evaluating potential risks on a 5x5 Probability-Impact Matrix. Risk Severity is calculated as Severity = Likelihood (1–5) x Impact (1–5), categorizing risks into Low (1–6), Medium (8–12), and High (15–25).

ID	Risk Description & Event	Lik	Imp	Sev	Mitigation Strategy & Contingency Plan
R-01	Cross-Contamination of Bins (e.g., food in blue bin)	4	4	16	High-contrast bilingual pictorial signage; foot-pedal color coding; daily morning inspection.
R-02	Student & Staff Apathy / Resistance to Segregation	4	3	12	Student awareness campaign; placement of peer green volunteers during initial week.
R-03	Foul Odor & Fruit Fly Infestation in Wet Bin	3	4	12	Tight-fitting foot-pedal lid; daily clearance; spray of diluted citronella/bio-enzyme.
R-04	Anaerobic Putrefaction & Odor in Pilot Composter	3	3	9	Maintain 30:1 C:N ratio using crushed dry leaves; turn compost twice weekly for aeration.
R-05	Vandalism, Mishandling, or Station Displacement	2	3	6	Station anchored to heavy timber skid; placed directly opposite faculty room CCTV.
R-06	Paint Chipping or Peeling on Plastic Buckets	3	2	6	Surface pre-sanded with 120-grit emery paper; anti-rust primer applied prior to enamel.
R-07	Failure or Delay in AMC Scrap Recycler Handover	2	4	8	Pre-negotiated MoU with licensed local kabadiwala/AMC aggregator; secure dry storage depot.
R-08	Monsoon Rain Waterlogging & Corrosion	2	3	6	Overhead sloped canopy (5° slope); base skid elevated 150 mm on anti-corrosive rubber pads.
R-09	Injury or Pathogen Exposure to Waste Handlers	2	4	8	Mandatory heavy-duty puncture-resistant nitrile gloves, N95 masks, and disinfectant.
R-10	Budget Overrun due to Consumable Purchases	2	2	4	Strict 10% contingency reserve allocated; bulk purchases from verified wholesale vendors.

Chapter 17
SAFETY AND ENVIRONMENTAL CONSIDERATIONS

Solid waste management projects involve inherent biological, chemical, and ergonomic hazards that necessitate stringent occupational health and safety protocols alongside environmental protection standards.

17.1 Occupational Health & Personal Protective Equipment (PPE)

Waste handling carries risks of lacerations from broken glass or staples, contact with pathogenic microorganisms, and inhalation of dust or fungal spores. The team instituted the following non-negotiable safety protocols:

- 1. Mandatory Puncture-Resistant Gloves: All team members and housekeeping staff must wear heavy-duty puncture-resistant nitrile gloves during waste collection, segregation auditing, and weighing. Disposable medical latex gloves are strictly prohibited due to tear vulnerability.
- 2. Respiratory Protection: N95 particulate respirators must be worn during composter turning and gravimetric waste sorting to filter airborne bio-aerosols, dust, and microbial spores.
- 3. Hygiene & Hand Sanitation: Strict prohibition of open hand sorting. Hand tongs or blunt scrapers must be used when separating commingled items during daily audits. Hands must be washed with antibacterial soap immediately following activities.

17.2 Environmental Safeguards & Vector Management

- 1. Leachate Containment: The pilot composter is elevated on a concrete catch tray lined with sawdust to capture any incidental leachate, preventing liquid runoff into campus drainage or soil.
- 2. Vector & Vermin Control: To prevent mosquitoes, houseflies, and rodents from colonizing bins, all lids feature airtight lip seals, and the wet bin is cleared daily by 4:00 PM prior to evening building closure.
- 3. Zero Open Burning: Strict adherence to Solid Waste Management Rules 2016 Section 15(g), completely prohibiting the open burning of waste, fallen garden foliage, or plastic packaging anywhere on college grounds.

Chapter 18
ADVANTAGES AND ENVIRONMENTAL BENEFITS

The implementation of a decentralized solid waste segregation and recycling system on an engineering campus yields quantifiable ecological, financial, and pedagogical advantages.

18.1 Quantitative Environmental Impact Metrics

- 1. High Landfill Diversion Rate (88.5%): During the 14-day pilot audit, the single station collected an average of 7.8 kg/day of waste. By separating 4.1 kg/day of wet waste for composting and 2.8 kg/day of dry recyclables, exactly 6.9 kg/day was diverted away from municipal dumpsites—achieving an empirical Landfill Diversion Rate of 88.5%.
- 2. Greenhouse Gas Abatement: When organic waste decomposes anaerobically in unsegregated landfills, it emits methane (CH4), a potent greenhouse gas with a global warming potential 28 times greater than CO2. Aerobic composting of 4.1 kg/day of food waste prevents approx. 4.9 kg of CO2-equivalent emissions daily (approx. 980 kg CO2e over an academic year per station).
- 3. Natural Resource Conservation: Every 54 kg of paper collected and sent for recycling conserves approximately one mature tree, 1,300 litres of water, and 160 kWh of electrical energy compared to manufacturing virgin paper pulp.
- 4. In-Situ Soil Conditioning: The pilot composter converts food scraps into nutrient-dense, stabilized organic humus, returning vital nitrogen, phosphorus, and potassium (NPK) to campus garden beds without chemical fertilizers.

18.2 Economic and Institutional Benefits

- 1. Frugal Capital Expenditure: Reclaiming 20L commercial paint drums and scrap timber produced a fully functional station for ₹1,670 INR—delivering a 91% cost saving compared to commercial fiberglass bins priced at ₹18,500.
- 2. Recyclable Monetization: Clean segregated paper, cardboard, and PET bottles command premium scrap market rates (₹12–15/kg for paper, ₹22–28/kg for PET). Accumulated funds can be channeled directly into the student Green Club to sponsor campus tree plantation.
- 3. Institutional Accreditations: Elevates L. D. College of Engineering's institutional rating under Swachh Campus national rankings, NAAC criteria, and National Institutional Ranking Framework (NIRF) sustainability parameters.

Chapter 19
LIMITATIONS AND FUTURE IMPROVEMENTS

In accordance with rigorous academic scholarship, the limitations encountered during the pilot execution are candidly analyzed alongside a phased roadmap for future enhancement.

19.1 Observed Project Limitations

- 1. Limited Volumetric Capacity: The 20-litre bucket capacity requires daily clearing; during high-intensity events (such as department symposiums or exams), the blue recyclable bin experiences rapid volumetric overflow from large cardboard boxes.
- 2. Manual Data Entry: Daily logging relies on manual weighing using a handheld digital scale and Google Forms data entry, creating vulnerability to human omission or recording delays.
- 3. Behavioral Friction: While student compliance reached 92% by Day 14, initial days exhibited contamination from greasy food wrappers tossed into the blue paper bin, indicating that behavioral inertia requires persistent nudges.
- 4. Outdoor Weather Protection: The prototype was deployed in an indoor hallway corridor; outdoor deployment would require enhanced UV-stabilized polycarbonate roofing and anchor bolting against monsoon gusts.

19.2 Roadmap for Future Technical Improvements

- 1. IoT-Enabled Fill-Level Telemetry: Integrate an ultrasonic distance sensor (HC-SR04) and ESP32 microcontroller inside each bin lid to measure bin fill level in real time. When the bin reaches 80% capacity, an automated alert is triggered via Wi-Fi to campus housekeeping staff.
- 2. Gamified Student Mobile App: Develop a cross-platform mobile application ('Green LDCE') allowing students to scan bin QR codes when disposing of clean recyclables to earn academic green credits redeemable for library perks or canteen discounts.
- 3. Solar-Powered Rotary Composter: Transition from the 5L pilot compost bin to a centralized 100-litre dual-chamber rotating composting drum installed near the campus botanical garden, powered by a small solar PV aeration fan.

Chapter 20
LEARNING OUTCOMES

The execution of this Project Management Micro-Project provided invaluable experiential learning, enabling B.E. Semester V Information Technology students to bridge abstract management theory with real-world engineering problem-solving.

Learning Domain	Theoretical Concept Applied	Demonstrated Engineering Competency
Project Management Frameworks	PMBOK 7th Ed, WBS, Gantt Scheduling, Critical Path Method (CPM).	Formulated 3-level WBS, sequenced 10 activities, mapped dependencies, and controlled 8-week timeline.
Cost Engineering & Budgeting	Itemized Cost Estimation, Contingency Reserves, Cost-Benefit Analysis.	Delivered complete station at ₹1,670 INR vs ₹18,500 commercial baseline, staying within budget.
Risk Management	Probability-Impact Scoring, Risk Register, Contingency Trigger Planning.	Identified 10 operational risks, enforced PPE protocols, and achieved zero safety incidents.
Information Technology Integration	Google Cloud Forms, Sheets formulas, QR code tagging, Web Dashboards.	Engineered automated data collection pipeline tracking daily waste inflow and diversion metrics.
Environmental & Civic Engineering	SWM Rules 2016, Circular Economy, Aerobic Composting C:N ratios.	Achieved 88.5% landfill diversion rate and established formal recycler supply chain.

Chapter 21
CONCLUSION

This Project Management Micro-Project successfully demonstrates that sustainable solid waste management on a large engineering college campus does not require capital-intensive machinery or exorbitant expenditure. By applying core project management methodologies—structured Work Breakdown Structures, rigorous Gantt scheduling, unambiguous RACI matrices, proactive risk registers, and disciplined cost control—a 5-member student team engineered, fabricated, and deployed an effective 3-stream waste segregation station within 8 weeks and under a budget of ₹1,670 INR.

The empirical findings from our 14-day pilot deployment in the IT Department corridor validate the system's operational efficacy. The station diverted 88.5% of generated waste away from municipal landfills, separating 4.1 kg/day of wet organic matter for composting and 2.8 kg/day of clean dry recyclables for authorized reprocessing. Student segregation accuracy climbed from an initial 54% to 92% through clear bilingual visual signage, foot-pedal hands-free hygiene, and digital audit monitoring.

The project stands as a reproducible, low-cost model that can be scaled across all 14 engineering departments at L. D. College of Engineering and adapted by academic institutions across Gujarat. In doing so, it embodies the spirit of the Swachh Bharat Mission, Mission LiFE, and statutory Solid Waste Management Rules 2016, proving that engineering students can lead the transition toward circular, zero-waste campuses.

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